

AMD DEVELOPER HACKATHON · ACT II · TRACK 1

ZeroProof

A **zero-token, proof-carrying** routing agent. Every task is answered by the **cheapest source that can be trusted** — a deterministic exact solver when the category admits proof, and a bundled small local LLM for the linguistic ones — at **0 Fireworks tokens**.

0 Fireworks tokens

proof-carrying answers

built to survive the refresh

How Track 1 is actually won

80%

Accuracy gate

An LLM judge scores intent on a **hidden, refreshed** task set generated *after* submission.

↓ tokens

Then rank by tokens

Among those who clear the gate, ranking is **ascending total Fireworks tokens**. Zero is #1.

tie → acc

Tie-break

Equal tokens $\approx$ generalized accuracy. So we maximize accuracy in the 0-token cohort.

The winning play is not subtle: **score exactly zero tokens, clear the gate with margin, and hold the highest generalized accuracy in the 0-token cohort.**

THE INSIGHT EVERYONE MISSED

0 tokens is **table stakes** — not a moat

The whole field already converged on “local model + verify + escalate, aiming for 0 tokens.” Dozens of teams sit at 0 on the public board. So zero tokens no longer separates anyone.

The real, hidden battle

Generalized accuracy in the 0-token cohort on the **refreshed** set — where overfit entries collapse. A real team scored **>80% locally** → **~52% officially**.

ZeroProof spends its **entire engineering budget** on that one decisive variable.

WHY WE WIN

The three-part moat

1 · Algorithm

The widest **exact-solver** coverage in the field. A correct algorithm is **paraphrase-invariant** — the refresh cannot break it.

2 · Verification

Proof-carrying answers. Code is executed, math re-derived, logic constraint-solved, entities rule-corrected. A wrong “free” answer is structurally rejected.

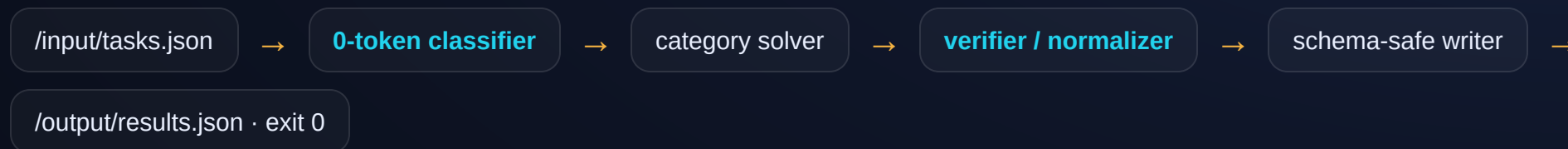
3 · Measurement

The **Proving Ground** — a generalization engine that optimizes the hidden-set score competitors can't see. Uncopyable edge.

Idea and architecture set the ceiling. The Proving Ground, run relentlessly, is how we actually reach it.

ARCHITECTURE · ALL ON CPU, 0 TOKENS

Route to the cheapest trustworthy source



Provable → exact solvers

math (symbolic · arithmetic-chain · executed program-of-thought), code (execute tests + repair), logic (constraint solver), NER (spaCy + rule correction). **Proof-carrying.**

Linguistic → local LLM, governed

factual (self-consistency medoid), sentiment (contrastive-cue governance), summarization (deterministic count/word-cap enforcement). Bundled 1.5B GGUF, **0 tokens**.

A default-OFF Fireworks valve exists in code only — an audit-safe, factual-only insurance lever. The win path never touches it.

PER-CATEGORY STRATEGY

Eight categories, every answer earned

CATEGORY	PATH	HOW IT'S PROVEN	TOKENS
Mathematical reasoning	deterministic + PoT	symbolic / operation-chain / executed program; samples agree	0
Code generation	LLM + execution	run against parsed/derived tests; repair loop to green	0
Code debugging	LLM + execution	fix → re-execute to green	0
Logical / deductive	constraint solver	unique answer across all satisfying models	0
Named-entity recognition	spaCy + rules	gazetteer / acronym / person-verb corrections	0
Sentiment analysis	LLM + rules	contrastive-cue governance; both-sides reason enforced	0
Text summarization	LLM + enforcement	exact sentence/bullet counts & word caps	0
Factual knowledge	local LLM	self-consistency medoid (honest top risk)	0

We optimize the score competitors can't see

MEASURE

Generate unseen **paraphrase / harder / adversarial** variants with ground truth. Judge each, meter tokens, simulate 2 vCPU / 4 GB / 10 min / 30 s, run a hardcode guard.

DIAGNOSE → IMPROVE

Rank misses by hidden-set cost; make one targeted fix; keep only if generalized accuracy rises with **zero regression**. Bank each best.

ITERATION	CHANGE	DET ACC	OVERALL
v1 baseline	initial solvers	88.3%	—
v2	gerund/comma parsing · ratio solver · context-aware sentiment	98.3%	—
v3	chain start-fallback	100%	—
v6	deterministic-first + per-task budget (real 1.5B model)	100%	95.3%

CHAMPIONSHIP-LOCKED

Measured, not asserted

100%

Deterministic

math · logic · NER ·
sentiment · summarization,
on unseen variants
(400/400).

95.3%

Overall

including the bundled 1.5B
model (factual + code).

0

Fireworks tokens

proven by the token meter.



Constraints green

slowest task 13.8 s (< 30 s)
· ~2.0 GB RAM · image
~1.6 GB.

All numbers come from **paraphrased / unseen generated variants**, never the public sample tasks — a genuine anti-overfit signal. The hardcode guard is clean.

WHY THIS WINS RANK #1

When the refresh knocks everyone down, ZeroProof stays on top.

Compliant & pure

0 tokens · no external calls · no hardcoded answers · no secrets · ≤ 5 GB · license-clean (MIT + Apache-2.0 model).

Robust by construction

Correct algorithms don't care how a prompt is worded. Wrong free answers are rejected before they ship.

Overfit “100%” entries crater on the refresh. ZeroProof's margin is **earned** on unseen variants — the one thing that carries through to the final score.